

Architecture Justification

Alumni Mentorship & Mock Interview Platform

Architectural Choice: Layered Architecture

The Alumni Mentorship & Mock Interview Platform uses a Layered Architecture consisting of a Presentation Layer, Business Layer, and Data Layer. The Presentation Layer handles interactions with students and alumni mentors, the Business Layer manages mentor profiles, availability, session scheduling, mock interviews and feedback, and the Data Layer manages persistent platform data. The Lab 3 handout identifies Layered Architecture as an option that provides clear separation between presentation, business logic, and data responsibilities.

Reason 1 – Separation of Responsibilities

The platform contains different responsibilities such as mentor profile management, session scheduling, mock-interview management, and feedback/scorecard handling. Layered Architecture separates these responsibilities into appropriate layers, making the system easier to understand, maintain, test, and modify without unnecessarily affecting other layers.

Reason 2 – Suitable for the Platform's Requirements

The platform needs to manage interactions between Student Mentees, Alumni Mentors, business logic, and stored information. The layered structure provides a clear flow from user requests in the Presentation Layer, through business operations such as mentor matching and scheduling, to the Data Layer where profiles, bookings, availability, and interview feedback are stored. This organization suits the functional requirements of the platform.

Security Advantage

Layered Architecture allows access to sensitive information such as student profiles, alumni information, bookings, and interview scorecards to be controlled through the Business Layer before reaching the Data Layer. This helps prevent direct and unauthorized access to stored data.

Performance Benefit

The separation of responsibilities allows frequently used operations, such as mentor profile retrieval, mentor matching, and availability checking, to be optimized within the appropriate business and data components without requiring changes to the

Presentation Layer. This can help maintain responsive system operation as usage increases.

Conclusion:

Layered Architecture is therefore selected because it provides a clear structure for separating the platform's user interface, business functionality, and data management while supporting its security and performance requirements.